Olabode T. Ajenifujah

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Research Interests

Verified AI agents for advanced manufacturing: tool-using and multi-agent large language model (LLM) systems that operate validated process models, with code-graded, pre-registered evaluation. Process development, in situ monitoring and qualification for additive manufacturing (laser powder bed fusion). Machine learning and multiphysics simulation for additive manufacturing, welding and semiconductor process equipment. Reliability of AI decision support in production.

Education

Carnegie Mellon University, Pittsburgh, PA Sep 2021 – Dec 2025

- Ph.D., Mechanical Engineering. Advisor: Prof. Amir Barati Farimani
- Thesis: *Machine Learning Models for Spatter Reduction in Additive Manufacturing*
- GEM Fellowship (2021)

University of Michigan, Ann Arbor, MI Sep 2014 – May 2016

- M.S.E., Chemical Engineering

University of Michigan, Ann Arbor, MI Jan 2009 – May 2014

- B.S.E., Chemical Engineering

Postdoctoral Research

Postdoctoral Research Fellow

University of Michigan, Mechanical Engineering | Advisors: Prof. Chenhui Shao and Prof. Jeffrey Abell Mar 2026 – Present

- Lead University of Michigan researcher on the General Motors Cognitive Manufacturing project: LLM agents that operate validated quality and maintenance models for ultrasonic metal welding (UMW).
- Built a process-agnostic evaluation harness that wraps validated diagnostic models as tools, grades every answer by exact match against a key computed in code (no LLM judge), and replays every tool call to locate each failure.
- Led a series of studies, several pre-registered, on where agents fail, hosted versus local open-weight models, grounded agent design, cost-aware quality decisions, and single- versus multi-agent designs for a multi-line plant.
- Co-developed a physics-structured neural ODE model of arc molten pools with workpiece thermal memory, learning pool states from process camera images with one network design for GTAW, GMAW and WAAM (second author).
- Sheet-metal stamping (with Y. Cao): explainable monitoring from press tonnage signals, including drift-aware detection, regional PCA that localizes anomalies to phases of the stroke, expert-verified root causes, physics-informed and language-guided synthetic faults, and fault-family classification.

Ph.D. Research

AgentAM: an AI agent for spatter- and defect-aware process optimization in LPBF

- Built an LLM agent that operates CFD-trained surrogates of spatter and melt-pool geometry for SS316L through the Model Context Protocol (MCP), maps the power–velocity window, assigns defect regimes and finds low-spatter, defect-free parameters with a margin-aware optimizer; evaluated on a 145-prompt benchmark with blinded rating.

Spatter mechanism and process window in laser powder bed fusion (LPBF)

- Combined multiphysics simulation and explainable machine learning to classify ejected spatter against the melt pool (94–96% accuracy) and to build data-driven process maps linking power and velocity to spatter

volume.

- Validated the high-fidelity OpenFOAM model against measured melt-pool dimensions, temperature and spatter size and velocity; transferred the classifier to FLOW-3D fields to map 281 power–velocity conditions about 18 times faster (about 4 versus 72 hours per case); identified recoil pressure, surface tension, pressure and vertical velocity as main drivers by SHAP analysis. Sponsored by the Army Research Laboratory.

Constitutive models for melt-pool features and spatter

- Developed machine-learning and symbolic-regression models of melt-pool length, width, depth, area and volume ($R^2 > 0.97$) and of spatter generation ($R^2 \approx 0.88$) from experimentally calibrated FLOW-3D and OpenFOAM simulations over 281 power–velocity cases.

Collaborative work: LLMs and generative models for additive manufacturing

- Co-developed AMGPT, a retrieval-augmented LLM for evidence-grounded querying of the metal AM literature.
- Co-developed a generative diffusion framework that reconstructs high-fidelity melt-pool temperature fields from low-fidelity simulations at a fraction of the computational cost.

Publications

Journal articles

- Khan, M. M., Bashir, S., Khan, M. W., & **Ajenifujah, O. T.** (2025). Reducing emissions and fuel consumption in supersonic aviation with ammonia hybrid engines. *International Journal of Hydrogen Energy*, 162, 150540.
- Ogoke, F., Liu, Q., **Ajenifujah, O.**, Myers, A. J., Quirarte, G., Beuth, J., Malen, J., & Barati Farimani, A. (2024). Inexpensive high fidelity melt pool models in additive manufacturing using generative deep diffusion. *Materials & Design*, 113181.
- Chandrasekhar, A., Chan, J., Ogoke, F., **Ajenifujah, O.**, & Barati Farimani, A. AMGPT: a large language model for contextual querying in additive manufacturing. *Additive Manufacturing Letters*, in press.
- **Ajenifujah, O. T.**, Nouralishahi, A., Carl, S., Eady, S., Zhao, J., & Thompson, L. T. (2021). Platinum supported on early transition metal carbides: efficient electrocatalysts for methanol electro-oxidation reaction in alkaline electrolyte. *Chemical Engineering Journal*, 406, 126670.
- Djire, A., Pande, P., Deb, A., Siegel, J., **Ajenifujah, O. T.**, He, L., Sleightholme, A. E., Rasmussen, P., & Thompson, L. T. (2019). Unveiling the pseudocapacitive charge storage mechanisms of nanostructured vanadium nitrides using in-situ analyses. *Nano Energy*, 60, 72–82.
- Djire, A., **Ajenifujah, O. T.**, & Thompson, L. T. (2018). Extent of pseudocapacitance in high-surface area vanadium nitrides. *Batteries & Supercaps*.
- Djire, A., Siegel, J. B., **Ajenifujah, O.**, He, L., & Thompson, L. T. (2018). Pseudocapacitive storage via micropores in high-surface area molybdenum nitrides. *Nano Energy*, 51, 122–127.
- Bazyari, A., Mortazavi, Y., Khodadadi, A. A., Thompson, L. T., Tafreshi, R., Zaker, A., & **Ajenifujah, O. T.** (2016). Effects of alumina phases as nickel supports on deep reactive adsorption of (4,6-dimethyl)dibenzothiophene: comparison between γ, δ, and θ-alumina. *Applied Catalysis B: Environmental*, 180, 312–323.
- Djire, A., **Ajenifujah, O. T.**, Sleightholme, A. E., Rasmussen, P., & Thompson, L. T. (2015). Effects of surface oxygen on charge storage in high surface area early transition-metal carbides and nitrides. *Journal of Power Sources*, 275, 159–166.
- Schaidle, J. A., Schweitzer, N. M., **Ajenifujah, O. T.**, & Thompson, L. T. (2012). On the preparation of molybdenum carbide-supported metal catalysts. *Journal of Catalysis*, 289, 210–217.

Under review and preprints

- **Ajenifujah, O. T.**, Ogoke, F., Wirth, F., Beuth, J., & Barati Farimani, A. Integrating multi-physics simulations and machine learning to define the spatter mechanism and process window in laser powder bed fusion. *Additive Manufacturing*, under review. arXiv:2405.07823.
- **Ajenifujah, O. T.**, & Barati Farimani, A. AgentAM: an AI agent for spatter- and defect-aware process optimization in laser powder bed fusion. Under revision. arXiv:2611.05171.

Manuscripts in preparation (University of Michigan and General Motors)

- Cao, Y., **Ajenifujah, O.**, Yang, Y., Zhang, Y. M., Abell, J., & Shao, C. Process-agnostic modeling of arc-based molten pools: physics-structured latent dynamics with workpiece thermal memory. In preparation.
- **Ajenifujah, O. T.**, et al. Orchestration reliability and failure modes of tool-using large language model agents for manufacturing quality and maintenance decision support. To be submitted to NAMRC 55.
- **Ajenifujah, O. T.**, et al. Hosted versus local open-weight LLM agents for manufacturing decisions: correctness, grounding, interface effects, and cost. In preparation for *Journal of Manufacturing Systems*.
- **Ajenifujah, O. T.**, et al. Designing grounded LLM agents for manufacturing decisions: evidence from ultrasonic metal welding. In preparation for *Journal of Manufacturing Systems*.
- **Ajenifujah, O. T.**, et al. From correct to right: language-model agents making compound quality decisions over validated ultrasonic welding models. In preparation for *Computers in Industry*.
- **Ajenifujah, O. T.**, et al. The role of work division, shared memory and verification in tool-using large language model agents for multi-line manufacturing quality decisions. In preparation.
- **Ajenifujah, O. T.**, Ogoke, F., Wirth, F., Beuth, J., & Barati Farimani, A. Derivation of constitutive model for melt pool features in additive manufacturing via machine learning. In preparation.
- **Ajenifujah, O. T.**, Ogoke, F., Wirth, F., Rollett, A., Beuth, J., & Barati Farimani, A. Discovery of spatter constitutive models in additive manufacturing using machine learning. In preparation.

Conference Presentations

- **Ajenifujah, O. T.**, Ogoke, F., Beuth, J., & Barati Farimani, A. "Application of multi-physics simulations and machine learning to predict spatter in laser powder bed fusion." TMS 2025, Las Vegas, NV, March 2025.
- **Ajenifujah, O. T.**, Ogoke, F., Wirth, F., Beuth, J., & Barati Farimani, A. "Combining multi-physics simulations with machine learning to elucidate spatter mechanisms and establish process map in laser powder bed fusion." MS&T 2024, Pittsburgh, PA, 2024.
- **Ajenifujah, O. T.**, Ogoke, F., Wirth, F., Beuth, J., & Barati Farimani, A. "Synergizing machine learning and multiphysics simulation for spatter process map generation in LPBF processes." Solid Freeform Fabrication Symposium (SFF 2024), Austin, TX, August 2024.
- Pak, P., Ogoke, F., Chandrasekhar, A., Guntuboina, C., **Ajenifujah, O.**, Rollett, A., Beuth, J., & Barati Farimani, A. "AMUI: a data-driven additive manufacturing user interface for process optimization." SFF 2024, Austin, TX, August 2024.
- Ogoke, F., Suresh, S., Liu, Q., Adamczyk, J., Bolintineau, D., **Ajenifujah, O.**, Myers, A., Quirarte, G., Garland, A., Heiden, M., Beuth, J., Malen, J., & Barati Farimani, A. "Applying generative deep learning models for cost-effective monitoring and simulation of LPBF processes." SFF 2024, Austin, TX, August 2024.
- **Ajenifujah, O. T.**, Eady, S., Nouralishahi, A., & Thompson, L. T. "Platinum supported on early transition metal carbides: efficient electrocatalysts for methanol electro-oxidation in alkaline electrolyte." NOBCChE National Conference, St. Louis, MO, November 2019. Oral.
- **Ajenifujah, O. T.**, Eady, S., Nouralishahi, A., & Thompson, L. T. "Effect of early transition-metal carbides as a support for low loading noble metals in direct ethanol fuel cells." NOBCChE National Conference, North Carolina, November 2016. Oral.

- **Ajenifujah, O. T.**, & Thompson, L. T. "Comparative phase study on tungsten carbides (α-WC and β-WC_{1-x}) as an electrocatalyst and as a support for low platinum loading in direct methanol fuel cells." Orlando, FL, September. Poster.
- Djire, A., **Ajenifujah, O. T.**, Sleightholme, A., Rasmussen, P., He, L., Siegel, J., & Thompson, L. T. "Charge storage mechanisms in carbides and nitrides." Electrochemical Society Meeting, Chicago, IL, May 2015. Oral and poster.
- Djire, A., **Ajenifujah, O. T.**, Sleightholme, A., Rasmussen, P., & Thompson, L. T. Four talks on charge storage in carbides and nitrides: AIChE Annual Meeting (Atlanta, November 2014; San Francisco, November 2013), Electrochemical Society Meeting (Cancun, October 2014; San Francisco, October 2013), and Power Sources Conference (Orlando, June 2014). Oral.

Research Funding and Support

General Motors Cognitive Manufacturing project (University of Michigan and General Motors) Mar 2026 – Present
- Postdoctoral support; lead University of Michigan researcher on AI agents for welding quality and maintenance.

Army Research Laboratory, Cooperative Agreement W911NF-20-2-0175 (PhD research on spatter and process windows in LPBF) Carnegie Mellon

GEM Fellowship, National GEM Consortium (PhD support) 2021

Teaching Interests

Materials processing and manufacturing processes; heat transfer; fluid mechanics; dynamics and controls; machine learning for mechanical engineers; new graduate course *Generative AI for Manufacturing* (14-week outline drafted); additive manufacturing process modeling.

Teaching and Mentoring

Teaching Assistant, Product Design (senior capstone)
Carnegie Mellon University Spring 2025
- Mentored capstone teams on design for manufacture and assembly, FMEA, life-cycle assessment and sustainability, with ABET-aligned documentation.
- Supported instruction on engineering ethics, project management, market research and professional communication.

Teaching Assistant, Mechanical Systems Experimentation
Carnegie Mellon University Fall 2023
- Led weekly labs for about 30 students on vibration analysis, system identification and feedback control (P, PI, PID) in MATLAB/Simulink.
- Held office hours and review sessions; graded reports with emphasis on experimental accuracy and uncertainty analysis.

Research Mentor, Undergraduate Research Opportunity Program (UROP)
University of Michigan 2026 – Present
- Mentor three first-year undergraduates on LLM agents for manufacturing (benchmark questions, LLM foundations).

Industry Experience

Computer Aided Engineer
Applied Materials Jan 2018 – Sep 2021
- Developed CFD and FEA models (CFD-ACE+, ANSYS, StarCCM+) for design optimization of epitaxy, atomic layer, metal deposition and oxide-etch chambers, including flow, thermal and reaction-chemistry analysis.
- Modeled film chemistry for GaN, AlN and oxide deposition, and ran illumination and ray-tracing simulations

(LightTools) of epitaxy lamp heads to optimize wafer heating.

- Built a SEMI S23 tool that reports total energy use and equivalent CO_2 for oxidation and metal deposition chambers.
- Compared five water-cooling designs for an epitaxy chamber.

Process Engineer
Applied Materials Jul 2017 – Jan 2018

- Improved chemical mechanical planarization (CMP) performance through post-polish cleaning chemistries.
- Characterized pad materials (DMA, Instron, tensiometer) and built a materials database for CMP pad design.

Leadership and Service

Associate applicant, African Studies Center (ASC)
University of Michigan 2026 – Present

- Applied for ASC Associate membership (executive committee decision March 2027); proposed focus on practical energy solutions for Africa's growing needs in manufacturing and health.

Founder and President
The Fujah Foundation (non-profit) Apr 2019 – Present

- Lead a non-profit that expands access to clean water and educational resources in underserved communities.
- Secured gifts and institutional partnerships; run school-supply, hygiene-kit and basic STEM enrichment programs; manage volunteers, governance, budgeting and impact tracking.

Vice President, Young Professional Network
Applied Materials Oct 2019 – Oct 2020

- Led program planning and member engagement; liaison to corporate executives and the executive sponsor; oversaw events, outreach and professional development committees.
- New-graduate recruiter for Applied Materials at the University of Michigan.

Skills

- **AI agents:** tool-using and multi-agent LLM systems, Model Context Protocol (MCP), local model serving (llama.cpp), OpenAI-compatible, Anthropic and Gemini APIs, agent evaluation design, trace-level failure analysis, pre-registered experiments
- **Machine learning:** explainable ML, symbolic regression, deep generative models, self-supervised learning, computer vision, anomaly detection
- **Programming:** Python, C, LaTeX
- **Simulation:** FLOW-3D, OpenFOAM, ANSYS (Fluent, Mesher, SpaceClaim), StarCCM+, CFD-ACE+, LightTools, SolidWorks
- **Laboratory:** catalyst synthesis and characterization (XRD, BET, TEM, ICP, TGA, chemisorption), electrochemistry (CV, EIS, LSV), packed-bed and batch reactors, gas chromatography, mechanical testing

Additional Research Experience

University of Michigan, Chemical Engineering
Graduate Student Research Assistant, Prof. Levi Thompson May 2015 – Jul 2017

- **Electrocatalysts for methanol and ethanol oxidation:** identified performance descriptors for Pt on six transition-metal carbides; developed synthesis routes for tungsten-carbide phases (γ-W_2C, β-WC, α-WC); bimetallic catalysts on carbide supports reached up to $2\times$ the activity of single-metal catalysts and $4\times$ commercial standards.
- **Electrochemical hydrogenation:** built and operated a PEM reactor to test carbides and nitrides for crotonaldehyde hydrogenation.
- **Supercapacitors:** identified vanadium carbide as a promising electrode; used cyclic voltammetry and SANS to study charge storage in Mo_2N and VN.

University of Michigan, Chemical Engineering
Undergraduate Research Assistant, Prof. Levi Thompson May 2011 – May 2015

- Removed oxygen passivation layers from carbides and nitrides, raising capacitance by 11% (Mo_2N) to 208% (W_2C); synthesized pure-phase carbides and nitrides.
- Studied Fischer–Tropsch and water–gas shift on molybdenum-carbide catalysts, and MOF and carbide adsorbents for diesel desulfurization.